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8TBS | pdb_00008tbs

Structure of human erythrocyte pyruvate kinase in complex with an allosteric activator AG-946

PDB DOI: https://doi.org/10.2210/pdb8TBS/pdb

Classification: TRANSFERASE/ACTIVATOR
Organism(s): Homo sapiens
Expression System: Escherichia coli
Mutation(s): No

Deposited: 2023-06-29 **Released:** 2023-12-27
Deposition Author(s): Jin, L., Padyana, A.
Funding Organization(s): Not funded

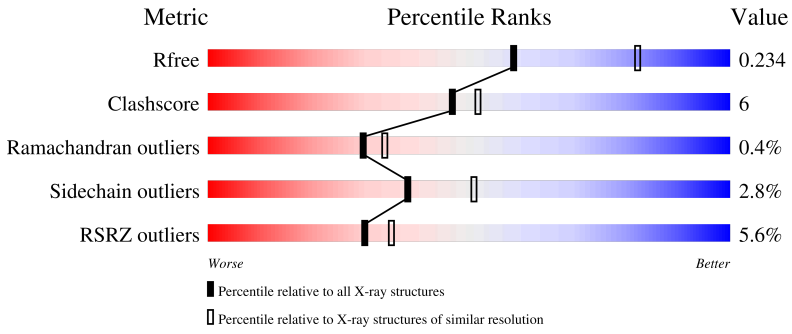
Experimental Data Snapshot

Method: X-RAY DIFFRACTION
Resolution: 2.35 Å
R-Value Free:
0.234 (Depositor), 0.234 (DCC)
R-Value Work:
0.193 (Depositor), 0.193 (DCC)
R-Value Observed:
0.195 (Depositor)

Starting Model: experimental
View more details

wwPDB Validation

3D Report | Full Report



Ligand Structure Quality Assessment



This is version 1.1 of the entry. See complete history.

Literature

Structure-Based Design of AG-946, a Pyruvate Kinase Activator.

Liu, T., Padyana, A.K., Judd, E.T., Jin, L., Hammoudeh, D., Kung, C., Dang, L.

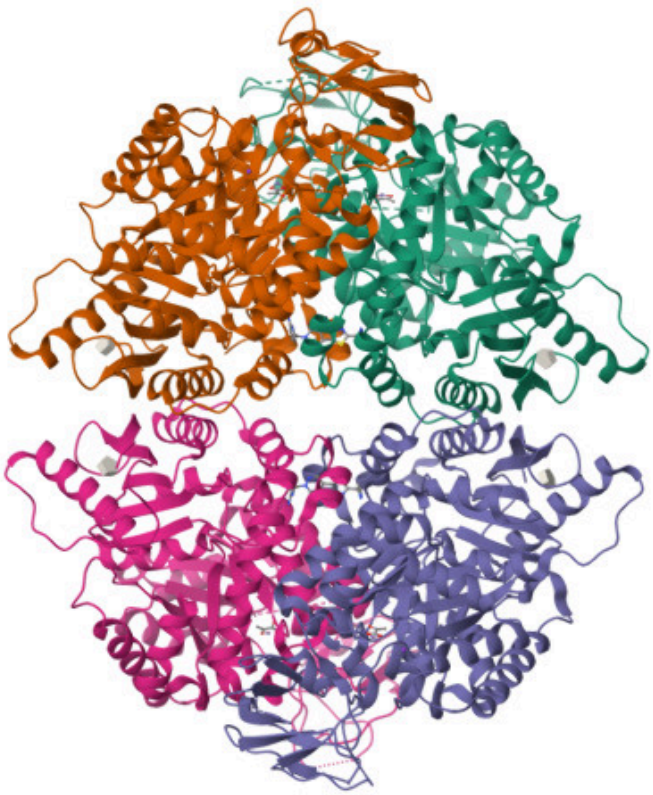
(2024) ChemMedChem **19**: e202300559-e202300559

PubMed: 38109501
DOI: https://doi.org/10.1002/cmdc.202300559
Primary Citation Related Structures:
8TBS, 8TBT, 8TBU

PubMed Abstract:
Pyruvate kinase (PK) is the enzyme that catalyzes the conversion of phosphoenolpyruvate and adenosine diphosphate to pyruvate and adenosine triphosphate in glycolysis and plays a crucial role in regulating cell metabolism. We describe the structure-based design of AG-946, an activator of PK isoforms, including r...
View More

Organizational Affiliation:
Ensem Therapeutics, 880 Winter St, Waltham, MA 02451, USA.
View More

Biological Assembly 1



Explore in 3D: Structure | Sequence Annotations | Electron Density | Validation Report | Ligand Interaction (HVI)

Global Symmetry: Dihedral - D2 (Explore in 3D)
Global Stoichiometry: Homo 4-mer - A4

Find Similar Assemblies

Biological assembly 1 assigned by authors and generated by PISA (software)

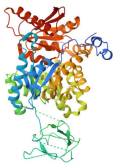
Macromolecule Content

- Total Structure Weight: 474.99 kDa
- Atom Count: 31,741
- Modeled Residue Count: 4,060
- Deposited Residue Count: 4,352
- Unique protein chains: 1

Macromolecules

Find similar proteins by: Sequence | 3D Structure

Entity ID: 1

Molecule	Chains	Sequence Length	Organism	Details	Image
Pyruvate kinase PKLR	A, B, C, D, E	544	Homo sapiens	**Mutation(s):** 0 **Gene Names:** PKLR, PK1, PKL **EC:** 2.7.1.40	

UniProt & NIH Common Fund Data Resources

Find proteins for P30613 *(Homo sapiens)* Explore P30613 Go to UniProtKB: P30613

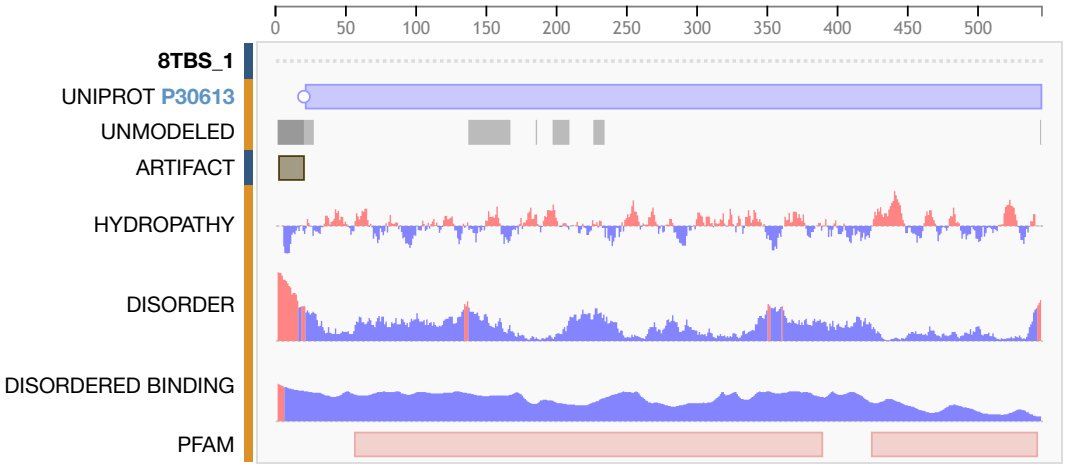
PHAROS: P30613 GTEx: ENSG00000143627

Entity Groups

Sequence Clusters	30% Identity, 50% Identity, 70% Identity, 90% Identity, 95% Identity, 100% Identity
UniProt Group	P30613

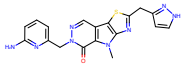
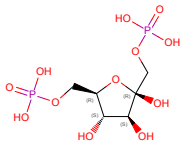
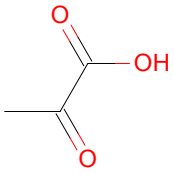
Sequence Annotations Expand

Reference Sequence 8TBS_1



Small Molecules

Ligands 5 Unique

ID	Chains	Name / Formula / InChI Key	2D Diagram	3D Interactions
HVI *(Subject of Investigation/LOI)* Download: Instance Coordinates	DA [auth F], M [auth B], MA [auth H], V [auth D]	**6-[(6-aminopyridin-2-yl)methyl]-4-methyl-2-[(1H-pyrazol-3-yl)methyl]-4,6-dihydro-5H-[1,3]thiazolo[5',4':4,5]pyrrolo[2,3-d]pyridazin-5-one** C_{18} H_{16} N_8 O S DIUOELXIXSCFCR-UHFFFAOYSA-N		Interactions Interactions & Density
FBP Download: Instance Coordinates		**1,6-di-O-phosphono-beta-D-fructofuranose** C_6 H_{14} O_{12} P_2 RNBGYGVWRKECFJ-ARQDHWQXSA-N		Interactions Interactions & Density
PYR Download: Instance Coordinates		**PYRUVIC ACID** C_3 H_4 O_3 LCTONWCANYUPML-UHFFFAOYSA-N		Interactions Interactions & Density
MN Download: Instance Coordinates		**MANGANESE (II) ION** Mn WAEMQWOKJMHJLA-UHFFFAOYSA-N	Mn^{+2}	Interactions Interactions & Density
K Download: Instance Coordinates		**POTASSIUM ION** K NPYPAHLBTDXSSS-UHFFFAOYSA-N	K^{+}	Interactions Interactions & Density

Experimental Data & Validation

Experimental Data

Method: X-RAY DIFFRACTION
Resolution: 2.35 Å
R-Value Free: 0.234 (Depositor), 0.234 (DCC)
R-Value Work: 0.193 (Depositor), 0.193 (DCC)
R-Value Observed: 0.195 (Depositor)

Space Group: P 2_1 2_1 2_1

Unit Cell:

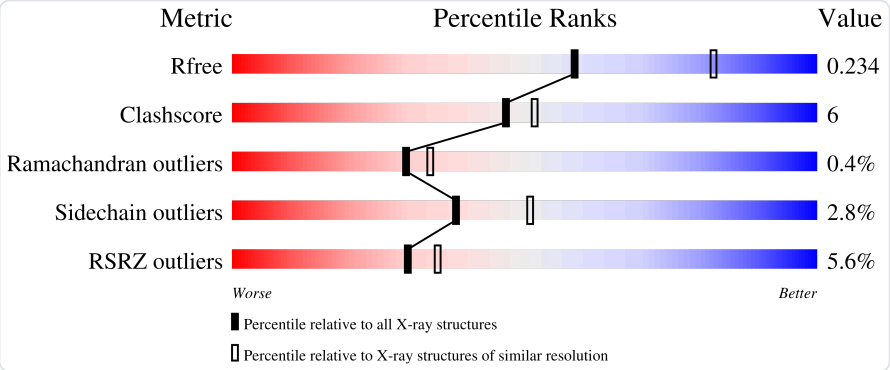
Length (Å)	Angle (°)
a = 110.32	α = 90
b = 122.596	β = 90
c = 378.458	γ = 90

Software Package:

Software Name	Purpose
PHENIX	refinement
HKL-2000	data reduction
HKL-2000	data scaling
PHASER	phasing

Structure Validation

View Full Validation Report



Ligand Structure Quality Assessment



Entry History & Funding Information

Deposition Data

Released Date: 2023-12-27
Deposition Author(s): Jin, L., Padyana, A.

Funding Organization	Location	Grant Number
Not funded		--

Revision History (Full details and data files)

- **Version 1.0: 2023-12-27**
  Type: Initial release
- **Version 1.1: 2024-03-20**
  Changes: Database references

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Partnerships & Collaborators:

RCSB PDB Core Operations are funded by the U.S. National Science Foundation (DBI-2321666), the US Department of Energy (DE-SC0019749), and the National Cancer Institute, National Institute of Allergy and Infectious Diseases, and National Institute of General Medical Sciences of the National Institutes of Health under grant R01GM157729. RCSB PDB uses resources of the National Energy Research Scientific Computing Center (NERSC), a Department of Energy User Facility.